you can always divide the distance between here and there or the duration between now and then in half and in half again, endlessly partitioning space and time into ever smaller units. Instead, when you get down to the Planck length (the length of a string) and Planck time (the time it would take light to travel the length of a string) and try to partition space and time more finely, you find you can't. The concept of "going smaller" ceases to have meaning once you reach the size of the *smallest* constituent of the cosmos. For zero-sized point particles this introduces no constraint, but since strings have size, it does. If string theory is correct, the usual concepts of space and time, the framework within which all of our daily experiences take place, simply don't apply on scales finer than the Planck scale—the scale of strings themselves.

As for what concepts take over, there is as yet no consensus. One possibility that jibes with the explanation above for how string theory meshes quantum mechanics and general relativity is that the fabric of space on the Planck scale resembles a lattice or a grid, with the "space" between the grid lines being outside the bounds of physical reality. Just as a microscopic ant walking on an ordinary piece of fabric would have to leap from thread to thread, perhaps motion through space on ultramicroscopic scales similarly requires discrete leaps from one "strand" of space to another. Time, too, could have a grainy structure, with individual moments being packed closely together but not melding into a seamless continuum. In this way of thinking, the concepts of ever smaller space and time intervals would sharply come to an end at the Planck scale. Just as there is no such thing as an American coin value smaller than a penny, if ultramicroscopic spacetime has a grid structure, there would be no such thing as a distance shorter than the Planck length or a duration shorter than the Planck time.

Another possibility is that space and time do not abruptly cease to have meaning on extremely small scales, but instead gradually morph into other, more fundamental concepts. Shrinking smaller than the Planck scale would be off limits not because you run into a fundamental grid, but because the concepts of space and time segue into notions for which "shrinking smaller" is as meaningless as asking whether the number nine is happy. That is, we can envision that as familiar, macroscopic space and time gradually transform into their unfamiliar ultramicroscopic counterparts, many of their usual properties—such as